SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

Solution Requirements - Script-Controlled ACL

Phase 2 · Requirement Analysis

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This document specifies the functional and non-functional requirements for the Script-Controlled ACL project, which restricts access to student Institution Details records based on the user's role and the Branch field value, implemented natively on the ServiceNow Now Platform (instance dev443713.service-now.com).

Functional Requirements

FR-No	Requirement
FR-1	The system shall store student Institution Details in a custom table (u_institution_details) with the fields Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.
FR-2	The Branch field shall be a choice field restricted to the values ECE, EEE, and CSE.
FR-3	The system shall enforce record-level access through four Access Control rules on u_institution_details, one each for read, create, write, and delete.
FR-4	The READ rule shall be Advanced, require the bb1 role, carry the data condition Branch is EEE, and run a script that returns true for admin and bb1 and false for everyone else.
FR-5	The CREATE, WRITE, and DELETE rules shall require the roles bb2, bb3, and bb4 respectively, with no condition or script.
FR-6	A user holding bb1-bb4 (the EEE User) shall see and manage only EEE-branch records and shall be able to create new records via the New button.
FR-7	A user without the required roles shall be denied access and shown the message "Security constraints prevent access to requested page".
FR-8	A user with the admin role (System Administrator) shall see all records regardless of Branch value.
FR-9	The default ACL role auto-generated when the table was created (u_institution_details_user) shall be removed so each rule depends only on its bb role.
FR-10	Access decisions shall be verifiable through Impersonation for the branch user, a no-role user, and the administrator.

Non-Functional Requirements

NFR-No	Category	Requirement
NFR-1	Security	Access shall be governed jointly by role and the Branch field value, and editing ACLs shall require security_admin elevation, preventing unauthorised rule changes.
NFR-2	Usability	Branch users shall see only their own records with no manual filtering, and the New, Update, and Delete controls shall appear only when their role permits the action.
NFR-3	Performance	ACL evaluation shall be handled natively by the Now Platform on each request with negligible added latency on the current dataset.
NFR-4	Reliability	The rules shall be deny-by-default, so any user or record not matching a rule is blocked, protecting record integrity.
NFR-5	Maintainability	Access logic shall use a short, readable server-side script (gs.hasRole) and standard ACL configuration, editable without any code deployment.
NFR-6	Scalability	The role-and-field pattern shall extend to additional branches (ECE, CSE) and larger record volumes without redesign.